

Aryan Mehesare

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Summary

Final year Computer Science student with hands-on experience in software development & cloud-native technologies. Built & deployed applications using various technology stacks with exposure to CI/CD pipelines & system design fundamentals. Passionate about building scalable, efficient, & reliable software while solving real-world problems.

Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering — CGPA: 8.81

Chennai, India

August 2023 – August 2027 (Expected)

The Bishop's Co-Ed School

Grade 12: 82.2% | Grade 10: 98.4%

Pune, India

July 2009 – May 2023

Experience

AI Engineer Intern

May 2026 – Jul 2026

SpeedTech.ai

Pune, India

- Built and deployed multimodal **MCP** integrations connecting **OpenAI** and **Google Gemini** models with audio-processing tools, enabling standardized real-time AI workflows in a telephony based scenario.
- Developed real-time speech-to-speech voice agents using the Azure Voice Live API and Google Gemini Live API, implementing session pooling that cut response latency by 30–40% and optimizing noise suppression and Voice Activity Detection (VAD).
- Designed a multi-agent CRM system on **LangGraph** and **LangChain** with a supervisor architecture dispatching to Analytics, Action, Report and Help sub-agents, integrated with **Azure OpenAI** and a semantic knowledge base.
- Implemented AI-driven lead qualification and follow-up automation with human-in-the-loop (HITL) interrupt flows, improving lead-handling efficiency by 4×.

Projects

Video Processing Microservices Platform

Apr 2026 – Jul 2026

- Architected a scalable, event-driven microservices platform using **Go**, **Python**, **Typescript** and **Kubernetes** for video processing.
- Reduced processing latency by **30%** by implementing a concurrent Go-based API Gateway and gRPC communication.
- Designed a robust message queue system using **RabbitMQ** to handle asynchronous tasks and prevent service bottlenecks.
- Implemented a **Redis** caching layer for metadata and job status, cutting database reads by **60%** & enabling real-time tracking.
- Tech Stack:** Python, Kubernetes, gRPC, Go, FastAPI, RabbitMQ, Redis, PostgreSQL, Typescript, Docker

Audio Deepfake Forensics

Dec 2025 – Feb 2026

- Engineered a dual-feature **ResNet-18** & **CNN** ensemble, achieving **3.31% EER** and **0.9940 ROC-AUC** on the ASVspoof dataset.
- Applied **Grad-CAM** & **Integrated Gradients** to visualize vocoder artifacts, validating defense against 13 unseen zero-day attacks.
- Developed a **Gradio** dashboard delivering real-time multimodal forensic analysis with **<22 ms** inference latency.
- Optimized generalization using **SpecAugment** masking and 80:20 score-level fusion, scaling detection precision to **99.98%**.
- Tech Stack:** PyTorch, Captum (XAI), Librosa, Gradio, Scikit-learn

SafeScape – IoT & ML Safety Platform

Sep 2024 – Nov 2024

- Engineered a real-time safety platform featuring geofencing and safe routing algorithms using Next.js and FastAPI.
- Deployed containerized services on **AWS EC2** using Docker & Nginx as a reverse proxy, separating web & ML inference workloads.
- Designed infrastructure to serve ML inference workloads (OpenCV, audio classification) alongside the web application.
- Tech Stack:** Next.js, FastAPI, PostgreSQL, Docker, AWS EC2, Nginx

Technical Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, Go, Bash

Web & Data: Next.js, React.js, Node.js, gRPC, RabbitMQ, Redis, MongoDB, PostgreSQL

DevOps: Docker, Kubernetes, Jenkins, Ansible, ArgoCD, Terraform, AWS (EC2, S3, IAM), Git, GitHub, GitLab CI, Linux, Nginx

Monitoring & Security: Prometheus, Grafana, Trivy, Snyk

Certifications & Achievements

- Oracle Cloud Infrastructure 2025 Generative AI Professional (1ZO-1127-25)
- Kubernetes Administration & Application Development - KodeKloud | AWS Cloud Architecture - Udemy
- Winner** — Sanrakshan Hackathon, IIT Bombay Techfest 2024
- Top 10** — Smart India Hackathon (SIH) Internal Rounds (2023, 2024)